

Solution Architecture

Date	29/09/2026
Team ID	UNM1425EduGenie
Project Name	EduGenie
Maximum Marks	5 Marks

ARCHITECTURE FLOW

PRESENTATION / CLIENT LAYER

HTML5 + CSS3 + JavaScript
EduGenie Web Interface

↓ ↑

API GATEWAY / SERVER

FastAPI
Uvicorn

↓ ↑

AUTH SERVICE

No separate authentication service in the current version
(Student authentication can be added as a future enhancement)

↓ ↑

CORE LOGIC SERVICE

EduGenie Learning Services

- Question Answering
- Topic Explanation
- Quiz Generation
- Content Summarization
- Learning Recommendations

↓ ↑

EXTERNAL API INTEGRATION

Google Gemini API

- AI-powered content generation
- Educational explanations
- Question answering
- Quiz generation
- Summarization
- Learning recommendations

↓ ↑

DATA / STORAGE LAYER

No persistent database in the current version

- Student input is processed dynamically
- AI-generated results are returned to the user

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the student-facing interface for selecting learning activities, entering content, generating responses, and viewing results.	HTML5, CSS3, JavaScript
API Gateway / Server	Receives HTTP requests from the frontend and routes them to the appropriate EduGenie service.	Python, FastAPI, Uvicorn
Auth Service	A separate authentication service is not implemented in the current version.	Not implemented
Core Logic Service	Processes the selected learning activity and prepares the appropriate request for AI generation.	Python, FastAPI
External API Integration	Provides Generative AI capabilities for educational responses.	Google Gemini API
Database / Storage	No persistent database is used in the current version. Requests and generated responses are processed dynamically.	Not applicable

Simple diagram to draw

